UNIT42 · FIRMWAREX

Firmware Security Assessment

MIPSEL controlled E2E firmware

Generated 10/20/2026, 11:01:06 AM

Executive summary

This assessment recorded **16 findings** across 2 audited binaries and 6 filesystem entries. Analysis status: **complete**.

Results describe the analysis performed on this image. They do not establish that the firmware is secure or that every candidate is exploitable. Severity and confidence are separate; no CVSS score is inferred.

Critical	High	Medium	Low	Info
0	1	6	9	0

Scope and coverage

Stage	Outcome	Restriction
extract	extracted	See scope limits below
static	completed	See scope limits below
emulation	completed	See scope limits below
symbolic	completed	See scope limits below
secrets crypto	completed	See scope limits below
normalize	completed	See scope limits below

Untrusted execution: blocked_pending_mentor_approval. Raw artifact access: disabled.

- Only registered controlled images execute; other images receive static and scanner analysis.
- Unknown path components are pseudonymized to prevent proprietary-string disclosure.
- Symbolic findings are bounded candidates for human review.

- Only executed stages contribute evidence. Blocked, skipped or incomplete analysis is not a clean bill of health.
- This is an automated firmware-image assessment, not a live-device or network scan.
- Recommendations are general review guidance, not verified fixes.

Binary protection summary

Binary	Architecture	NX	PIE	RELRO	Canary	FORTIFY
/bin/service-probe	MIPSEL	Not detected	Not detected	none	Present	Not detected
/bin/taint-demo	MIPSEL	Not detected	Not detected	none	Not detected	Not detected

Detailed findings

---

1. Command injection candidate High

Property	Result
CWE	CWE-78
Affected file	/bin/taint-demo
Architecture	MIPSEL
Confidence	probable
Review status	new
Address	0x500004
Function / sink	system
Detected	2026-10-20T05:30:51.308494Z

Evidence

Observation	Value
condition satisfiable	true

Observation	Value
Observed taint path	0x400680 → 0x500000 → 0x4006cc → 0x4006e8 → 0x500004
Reaching condition (diagnostic expression)	[{"op":"UGE","args":[{"op":"Extract","args":[247,240, {"symbol":"fwx_network_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args":[247,240, {"symbol":"fwx_network_0","bits":248}]}, {"constant":126,"bits":8}]},{"op":"UGE","args": [{"op":"Extract","args":[239,232, {"symbol":"fwx_network_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args … [excerpt; full expression in JSON export]
stack hash	e5d33603a1ee68b5b85a27213e73c3a8a502862e8f2cbfe90b4fc88e5bb6215b

Evidence reference: evidence:f01a238c3dc71b0754093b7d768a2230ad90bcbb673c362a1a59ee9c22a008e2. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review whether external input can reach command execution. Prefer structured APIs and strict allowlists; validate the candidate before remediation.

---

2. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/controlled.conf
Architecture	MIPSEL
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.435519Z

Evidence

Observation	Value
rule	credential
fingerprint	1daf3f2ef2e035e093b0f56dd085f850583031d172827d0aff4038a0e2afa4d4
byte offset	9
byte length	19

Evidence reference: evidence:817fc885639fd4f57e3b087d29ea1ae6bc40d4bf2cee2362fd95b340e9859636. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

3. Embedded secret candidate Medium

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	MIPSEL
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.435553Z

Evidence

Observation	Value
rule	credential

Observation	Value
fingerprint	f28635636b8c9687bb53be42780260da5e60fd146dda926c5a1635e341dedc76
byte offset	37
byte length	24

Evidence reference: evidence:62c426a8e11d0cab4f6f9d92f207e4f7c9f568804d263f30407daee17484508c. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

4. Embedded secret candidate Medium

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	MIPSEL
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.435574Z

Evidence

Observation	Value
rule	private_key
fingerprint	0d2707f8f67ee41af17ba6d1392ce94d1acfe2390cee1271e520eb96d3ad7cee

Observation	Value
byte offset	62
byte length	78

Evidence reference: evidence:8e9e07f442dae304624e4fa9c80cbefb592b166712904a93fa5101b6b8b77e97. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

5. Weak cryptography candidate Medium

Property	Result
CWE	CWE-328
Affected file	/etc/controlled.conf
Architecture	MIPSEL
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.435645Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	66cf0d996ec0021a2b98812603f716991a52abf3c1e63b9fcdf259bff7ce1dc7
byte offset	237
byte length	3

Evidence reference: evidence:e85245a818ce359986cf31e7fe15ff4ef3bf536d2a54185e6591e7d2e398e2a9. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

---

6. Weak cryptography candidate Medium

Property	Result
CWE	CWE-327
Affected file	/etc/controlled.conf
Architecture	MIPSEL
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.435670Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	2f258a413168040ae35beebe743631dd81b62b03d2100016d941c73db32ecd43
byte offset	255
byte length	3

Evidence reference: evidence:49390d2e525ffbd441e095f841e94d381911df6c846d96f1414a2aa94a1d871a. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

---

7. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/redacted-2d33e11295dbb335
Architecture	MIPSEL
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.435905Z

Evidence

Observation	Value
rule	credential
fingerprint	2539a753fa5615de28293ac5c62089db5d7194f089beb828f2805fdf3f9a723a
byte offset	38
byte length	25

Evidence reference: evidence:55fd6b30b66723f60d558fae25eef5a0b418aa82eaa449515a0f785ffd68e8eb. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

8. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/service-probe
Architecture	MIPSEL
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-10-20T05:30:47.814972Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

9. Embedded secret candidate Low

Property	Result
CWE	CWE-798

Property	Result
Affected file	/bin/service-probe
Architecture	MIPSEL
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.434711Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	4fd934ab27b3e4d5ffb1d8dfe420144aae11b9c04615dee6d983000b2ffa0c56
byte offset	470266
byte length	50

Evidence reference: evidence:4aa8e2356a09fb1b1e7170f17f4bfb54a35f4586d1e97b8fdeb1c729e824895f. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

10. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	MIPSEL

Property	Result
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.434773Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	ff2bbb54d6352d43b33bc5f573b791ed2ab9760d5391dc93f70eab9794f2fcab
byte offset	550141
byte length	26

Evidence reference: evidence:e5c9d4c92327e2d8a59cf3fc415772eeab0374e5eb4e5b4ebef2d71227ec00f4. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

11. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	MIPSEL
Confidence	heuristic
Review status	new

Property	Result
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.434800Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	8319af4b18cb01cbc027eb46104050965dd08e113b63515d417fcf6097c23b1e
byte offset	550173
byte length	26

Evidence reference: evidence:5f7e7e185c8dc8f19a050efc9e7765c489bec1a97fddb62e1b469160a3306c76. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

12. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	MIPSEL
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable

Property	Result
Detected	2026-10-20T05:30:51.434821Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	09b8ca00885179cab86e9ea749dd4d8cae7e216a3f3e8886f8d6d071c278a65a
byte offset	551040
byte length	36

Evidence reference: evidence:09a6103f3b45653db4d5ec0d61678222674358fc942b12d0ffadb04113819973. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

13. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	MIPSEL
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.434840Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	c6387ecc139038f582ede55c9c0926a747d3d3fd953ff825a8ac169c56ac7f3f
byte offset	551080
byte length	36

Evidence reference: evidence:9bdc2011a2271e72c60a64912f7d4d2b781c849bf4071927318cfe8295359581. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

14. Embedded secret candidate Low

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	MIPSEL
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.435592Z

Evidence

Observation	Value
rule	certificate

Observation	Value
fingerprint	47804325abedcb4b6c9d48e4983a9f816d9f26d155ebb168467f602395c22856
byte offset	141
byte length	75

Evidence reference: evidence:05b0b2127a3f2784f33baa916b4683ca1cebde89681fd4ddfa97cbe6cbbf15ea. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

15. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	MIPSEL
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.435609Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	80e26aa7e07c9409cee06092ec51db2ec463b3a5fa301a37f2ab4b16a285fc5d

Observation	Value
byte offset	29
byte length	32

Evidence reference: evidence:46d483f6bfe23b32f4fc5c884c161f5eab0b39d21a5bb33d608a5d629954ccd6. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

16. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	MIPSEL
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:51.435628Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	a5a18da06bd6af0b1cd3621f1da2d9cc942adc3d722ab5b56326e1571662d740
byte offset	259
byte length	29

Evidence reference: evidence:aa460d0b975b123fc26ccac0536b9fdb807dcf56719a830b361740526dcd3242. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

Assessment identity

Property	Value
Firmware ID	97365141-2b8a-46b5-8104-e9ce5ac5cf0f
SHA-256	6a1804f3f01b6ecb8f39e83ed73cdf52895f7e739a493f983c6ad0b4f9135902
Job ID	4d753927-0266-438d-a0de-5970ea128040
Image bytes	315392